Breast Cancer Classification Using Machine Learning and Feature Reduction

A Research Paper

Submitted by:
NAME: L. Snikitha Reddy
ROLL NO: 24B81A66Q1

Department of CSE

(AI&ML)

CVR College of Engineering

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Abstract and Keywords

Breast cancer diagnosis involves examining characteristics of breast tissue to determine whether a tumor is benign or malignant. This research proposes a machine learning approach for classifying breast tumors using numerical features derived from diagnostic measurements. The Wisconsin Diagnostic Breast Cancer dataset is used as a reproducible public dataset and contains 569 observations and 30 numerical features describing cell-nucleus characteristics.

The workflow includes data inspection, stratified train-test splitting, feature standardization, and dimensionality reduction using Principal Component Analysis (PCA). Logistic Regression, K-Nearest Neighbors (KNN), and Random Forest classifiers are trained and compared on the original standardized features and PCA-reduced features. Accuracy, precision, recall, F1-score, and confusion matrices are used for evaluation. PCA is fitted only on training data to avoid information leakage. Results are generated by executing the program rather than assuming fixed scores. The study demonstrates the relationship between dimensionality reduction and classification performance in a healthcare application. It is an educational experiment and is not a medical diagnostic tool.

Keywords: Breast cancer, machine learning, benign, malignant, PCA, feature reduction, Logistic Regression, KNN, Random Forest, classification metrics.

Introduction and Problem Statement

Breast cancer is a major health concern, and diagnostic assessment may involve examining the physical and microscopic characteristics of breast tissue. Machine learning methods can identify patterns in labeled datasets and use those patterns to classify new observations. Each row in a diagnostic dataset represents a sample, while each column represents a measured property.

The Wisconsin Diagnostic Breast Cancer dataset includes measurements such as radius, texture, perimeter, area, smoothness, compactness, concavity, symmetry, and fractal dimension. Multiple summaries are recorded for these characteristics, including mean values, standard errors, and worst values. Some features may be correlated, creating redundancy in the input space.

Logistic Regression provides a linear classification baseline, KNN predicts from nearby training examples, and Random Forest combines decision trees to model nonlinear relationships. Principal Component Analysis transforms correlated features into a smaller set of orthogonal components ordered by explained variance. PCA may reduce redundancy and computation, but it can also discard information useful for classification.

Problem Statement

Can breast tumors be classified as benign or malignant using machine learning, and can PCA reduce the number of input features while maintaining useful classification performance?

Objectives

1. Analyse a public breast cancer diagnostic dataset.
2. Inspect data structure, feature names, labels, and class distribution.
3. Split data into training and testing subsets using stratified sampling.
4. Standardize numerical features using training-set statistics.
5. Apply PCA and retain 95% of explained variance.
6. Develop Logistic Regression, KNN, and Random Forest classifiers.
7. Evaluate accuracy, precision, recall, and F1-score.
8. Examine confusion matrices.
9. Compare original and PCA-reduced features.
10. Discuss benefits and limitations of feature reduction.

Literature Review

Machine learning is widely investigated for medical classification because diagnostic datasets can contain complex relationships between numerical measurements and known outcomes. Classification algorithms learn patterns from labeled examples, but their performance depends on data quality, preprocessing, model assumptions, and evaluation methodology.

Logistic Regression is commonly used as a baseline for binary classification because it estimates class probabilities and provides a relatively straightforward decision boundary. It may be limited when relationships are nonlinear or inputs are highly correlated. Standardization is useful when feature magnitudes differ.

K-Nearest Neighbors classifies a sample using the labels of nearby training examples. Results depend on the distance measure, number of neighbors, and scale of each feature. Random Forest combines multiple decision trees and can represent nonlinear patterns and interactions. It can also provide feature-importance estimates, which should not be interpreted as proof of causation.

PCA reduces dimensionality by forming new features called principal components. Each component is a linear combination of the original features, and components are ordered by the variance they explain. Selecting a variance threshold provides a reproducible way to determine the reduced feature space.

Research Gap

A useful comparison should examine both the original features and the effect of reducing their dimensionality. PCA is unsupervised and selects components according to variance rather than target labels. Thus, a smaller feature set may simplify computation but may also lose information that helps distinguish benign from malignant samples.

This study compares three classifiers across original standardized features and PCA-reduced features. Multiple metrics and confusion matrices are used so that the effect of feature reduction is evaluated rather than relying on accuracy alone.

Research Scope

The work is limited to binary classification using the Wisconsin Diagnostic Breast Cancer dataset available through scikit-learn. It focuses on numerical features, standardization, PCA, and supervised classification. It does not include raw medical images, hospital patient records, treatment planning, or patient-specific advice.

1. Dataset Description

The dataset is available through the **sklearn.datasets.load_breast_cancer()** function. It is based on digitized images of fine needle aspirates of breast masses. The numerical measurements describe characteristics of cell nuclei.

- 569 observations.
- 30 numerical input features.
- Two target classes: malignant and benign.
- No missing values in the built-in dataset.
- Supervised binary classification task.

The original scikit-learn target encoding is 0 for malignant and 1 for benign. The program converts the target so malignant is the positive class. Therefore, the reported positive-class precision, recall, and F1-score focus on detecting malignant samples.

Feature family	Description	Examples
Size	Nucleus size measurements	Radius, perimeter, area
Texture	Variation in gray-scale values	Texture
Shape	Shape and contour	Compactness, concavity
Surface	Local boundary variation	Smoothness
Structure	Additional shape descriptors	Symmetry, fractal dimension

Dataset Summary

Property	Description
Dataset loader	sklearn.datasets.load_breast_cancer()
Samples	569
Input features	30
Target classes	Malignant and benign
Input type	Numerical
Learning task	Supervised binary classification
Feature reduction	PCA retaining 95% variance

The actual class counts can be inspected by running the program. Stratified splitting preserves approximate class proportions in the training and test subsets. A fixed random seed makes the experiment reproducible.

2. Data Preprocessing

Data preprocessing is important because raw data can contain categorical values, different numerical scales, or repeated observations. The selected built-in dataset is complete, and the experiment separates input features from target labels before splitting and model training.

Data Loading and Inspection

The dataset loader returns the feature matrix, target vector, feature names, and target names. The feature matrix is stored in a pandas DataFrame to support inspection. The program prints the dataset dimensions and original number of features.

Missing Value Analysis

The built-in dataset contains no missing values, so no imputation is required. If a different dataset file is used, missing values should be checked and handled before training.

Target Encoding

The original target value 0 denotes malignant and 1 denotes benign. The program transforms the target so malignant becomes 1 and benign becomes 0. This makes malignant-class recall and precision explicit in the evaluation.

Train-Test Split

The dataset is divided into 80% training data and 20% testing data. Stratification helps preserve the class proportions across the two subsets. The random_state parameter is set to 42 for reproducibility.

- 80% training data.
- 20% testing data.
- Stratified sampling.
- Random state = 42.

Feature Scaling

StandardScaler subtracts each feature's training-set mean and divides by its training-set standard deviation. It is fitted only on training data and then applied to training and testing data. This avoids allowing the test data to influence the transformation.

Scaling is especially important for KNN because distances may otherwise be dominated by variables with large numerical ranges. It is also useful for Logistic Regression and ensures PCA is not dominated merely by larger feature units.

Principal Component Analysis

PCA is applied after standardization. The setting n_components=0.95 retains enough components to explain at least 95% of variance in the training data. PCA is fitted on the standardized training set, and the same transformation is applied to the test set.

The number of retained components is determined by the data and printed at runtime. Explained variance is not identical to predictive information, so classification performance must still be measured.

Avoiding Data Leakage

The scaler and PCA transformation must not be fitted using the full dataset. Using test observations to fit preprocessing would allow information from the test set to influence the model evaluation. Fitting transformations on training data only gives a more appropriate estimate of performance on unseen data.

3. Proposed Methodology

Dataset Loading → Data Inspection → Target Preparation → Train-Test Split → Standardization → PCA → Model Training
→ Prediction → Evaluation → Model Comparison

Two feature representations are evaluated: all 30 standardized features and the reduced principal-component representation. The same split is used for both representations.

Logistic Regression

Logistic Regression is a supervised binary classification algorithm. It estimates the probability that a sample belongs to a class and uses a decision threshold to produce a label. It is included as a baseline because it is relatively simple and provides a useful comparison point.

The implementation sets max_iter=5000 to allow sufficient optimization iterations. The model is trained separately on original standardized features and PCA components.

K-Nearest Neighbors

K-Nearest Neighbors (KNN) predicts the class of a sample from the classes of its nearest training examples. This experiment uses five neighbors (K = 5). Since KNN depends on distance calculations, feature scaling is an important preprocessing step.

PCA changes the representation and distances between observations. Comparing KNN on both representations shows whether this dimensionality change affects its test-set performance.

Random Forest

Random Forest is an ensemble classifier combining predictions from multiple decision trees. Trees learn decision rules from training data, and combining them can improve robustness and represent nonlinear relationships. The implementation uses 200 trees and random_state=42.

The same configuration is trained on original standardized features and PCA-reduced features. Tree models do not generally require scaling in the same way as distance-based models, but the common feature-preparation workflow makes the comparison consistent.

Principal Component Analysis

PCA identifies directions in the standardized training data along which variance is largest. The first component explains the largest share of variance, and later components explain additional variance while remaining orthogonal to previous components.

PCA components are combinations of the original variables and are therefore less directly interpretable than measurements such as radius or area. Retaining 95% of variance preserves most of the variation in the input data, but does not guarantee that all target-relevant information is retained.

Evaluation Metrics

- Accuracy: proportion of all predictions that are correct.
- Precision: proportion of predicted malignant cases that are truly malignant.
- Recall: proportion of actual malignant cases correctly identified.
- F1-score: harmonic mean of precision and recall.
- Confusion matrix: counts correct and incorrect predictions by class.

The confusion matrix uses labels [malignant, benign]. Rows represent actual classes and columns represent predicted classes. This makes it possible to identify malignant samples incorrectly classified as benign.

4. Implementation

The system is implemented in Python using pandas and NumPy for data handling and scikit-learn for dataset loading, preprocessing, PCA, classification, and evaluation.

• Pandas — tabular data handling.

• NumPy — numerical operations.

• Scikit-learn — dataset, preprocessing, PCA, classifiers, metrics.

Step / algorithm	Configuration
Train-test split	80% train, 20% test; stratified
Random state	42
StandardScaler	Fit on training data only
PCA	95% explained variance
KNN	5 neighbors
Logistic Regression	Maximum iterations = 5000
Random Forest	200 trees

Complete Python Program

Run the following code in Google Colab or a Python environment with scikit-learn, pandas, and NumPy installed.

```
import pandas as pd
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection
import train_test_split from
sklearn.preprocessing import
StandardScaler from
sklearn.decomposition import
PCA
from sklearn.linear_model
import LogisticRegression
from sklearn.neighbors
import KNeighborsClassifier
from sklearn.ensemble import
RandomForestClassifier
from sklearn.metrics import
 (accuracy_score, precision_score,
 recall_score, f1_score, confusion_matrix,
 classification_report)

data = load_breast_cancer()
X = pd.DataFrame(data.data, columns=data.feature_names)
# Original: 0 malignant, 1
benign. Make malignant
positive. y = (data.target ==
0).astype(int)

print("Dataset shape:", X.shape)
X_train, X_test, y_train,
```

```
  y_test = train_test_split(
  X, y, test_size=0.20,
  random_state=42,
  stratify=y)

scaler = StandardScaler()
X_train_s =
scaler.fit_transform(X_t
rain) X_test_s =
scaler.transform(X_test)

pca = PCA(n_components=0.95)
X_train_p =
pca.fit_transform(X_trai
n_s) X_test_p =
pca.transform(X_test_s)
print("Original
features:", X.shape[1])
print("PCA components:",
pca.n_components_)
print("Variance retained:", pca.explained_variance_ratio_.sum()
```

7. Implementation (continued)

The remaining program trains each classifier using both feature representations and prints the measured metrics, confusion matrices, and final comparison table.

```
models = {
 "Logistic Regression":
 LogisticRegression(max_iter=5000), "KNN":
 KNeighborsClassifier(n_neighbors=5),
 "Random Forest":
    RandomForestClassifier(
    n_estimators=200,
    random_state=42)
}
results =
[] sets = [
 ("Original features", X_train_s,
 X_test_s), ("PCA features",
 X_train_p, X_test_p)
]
for feature_name, train_x,
   test_x in sets: for
   model_name, model in
   models.items():
      model.fit(train_x,
      y_train) pred =
      model.predict(test_x)
      results.append({
        "Feature Set":
        feature_name, "Model":
        model_name,
        "Accuracy": accuracy_score(y_test, pred),
        "Precision": precision_score(y_test, pred,
        zero_division=0), "Recall":
        recall_score(y_test, pred, zero_division=0),
        "F1-score": f1_score(y_test, pred, zero_division=0)
      })
      print("\\n", feature_name, "-", model_name)
      print(confusion_matrix(y_test, pred, labels=[1,
      0])) print(classification_report(
        y_test, pred, labels=[1, 0],
        target_names=["Malignant",
        "Benign"], zero_division=0))
print(pd.DataFrame(results).round(4).to_string(index=False))
```

Program Explanation

1. Load the dataset and store features in a DataFrame.
2. Convert the target so malignant is the positive class.
3. Split the data into stratified training and testing sets.
4. Fit StandardScaler on training features and transform both sets.
5. Fit PCA on training data and transform both sets.
6. Train each classifier on original and PCA features.
7. Calculate accuracy, precision, recall, and F1-score.
8. Print confusion matrices and the model comparison table.

5. Experimental Results

The models are evaluated on the same held-out test set. The experiment compares the original standardized features with the PCA-reduced feature representation. The Python program prints the measured values for each of the six model-feature combinations.

- Accuracy
- Malignant-class precision
- Malignant-class recall
- Malignant-class F1-score
- Confusion matrix

Test Set Results

Enter the values produced by the program in the following table. The values are not pre-filled because they must reflect actual execution of the experiment.

Feature set	Model	Accuracy	Precision	Recall	F1-score
Original	Logistic Regression	Output	Output	Output	Output
Original	KNN	Output	Output	Output	Output
Original	Random Forest	Output	Output	Output	Output
PCA	Logistic Regression	Output	Output	Output	Output
PCA	KNN	Output	Output	Output	Output
PCA	Random Forest	Output	Output	Output	Output

Interpreting the Results

Accuracy measures the overall proportion of correct predictions. Precision and recall refer to the malignant class because malignant is the positive label in the program. High recall indicates that a large share of actual malignant cases in the test set were detected, while precision indicates the reliability of malignant predictions.

Compare scores for each model before and after PCA. Dimensionality reduction may remove redundancy and reduce computational requirements, but it can also remove information useful for class separation. Conclusions should be based on the measured results.

Confusion Matrix

A confusion matrix shows correct and incorrect predictions for each class. The four outcomes are:

- True Positive (TP): malignant predicted as malignant.
- True Negative (TN): benign predicted as benign.
- False Positive (FP): benign predicted as malignant.
- False Negative (FN): malignant predicted as benign.

The program prints matrices in label order [malignant, benign]. The first row is actual malignant and the second row is actual benign. A false negative is a malignant sample predicted as benign and should be carefully examined in this educational setting.

	Predicted malignant	**Predicted benign**
Actual malignant	True Positive (TP)	False Negative (FN)
Actual benign	False Positive (FP)	True Negative (TN)

Original Features vs PCA Features

Comparison	**Original features**	**PCA features**
Input dimensions	30	Printed at runtime
Interpretability	Original measurements	Combinations of measurements
Redundancy	May include correlated inputs	May reduce correlated variation
Performance	Measured on test data	Measured on same test data

PCA should not automatically be described as better. If the scores remain similar with fewer components, the reduction may provide a compact representation. If scores decline, some useful information may have been lost.

6. Algorithm Comparison

Algorithm	Advantages	Limitations
Logistic Regression	Clear baseline; efficient; probability estimates	Linear boundary; affected by correlated inputs
KNN	Simple; captures local patterns	Sensitive to scaling, distance, and k
Random Forest	Models nonlinear patterns; ensemble-based	Less directly interpretable; more computation
PCA + classifier	Reduces dimensionality and redundancy	Components harder to interpret; may lose useful signal

Feature Reduction Analysis

PCA transforms the 30 original features into a smaller set of principal components. The components retain at least 95% of the variance in the standardized training data. Since PCA is unsupervised, the target labels do not influence component selection.

A compact representation can be beneficial when features are correlated or computation and storage matter. However, variance is not the same as predictive usefulness. A low-variance component may still help distinguish benign from malignant samples.

Model Selection

The best model should be selected using a stated criterion and actual evaluation results. Accuracy alone may not be sufficient. Malignant-class recall, precision, F1-score, and the confusion matrix should be considered together. The model that leads on one metric may not lead on all metrics.

Practical Significance

The experiment illustrates how preprocessing and feature reduction influence classification. It does not show that a model is ready for clinical use. Real-world use would require external validation, representative patient populations, bias and calibration assessment, and clinical oversight.

7. Discussion

This study compares Logistic Regression, KNN, and Random Forest on original standardized features and PCA-reduced features. The same stratified split is used for both representations. Fitting the scaler and PCA only on training data prevents the test set from influencing preprocessing.

The models have different characteristics. Logistic Regression provides a relatively simple linear baseline. KNN uses distances between samples, so scaling and dimensionality can influence its decisions. Random Forest combines tree-based rules and can represent nonlinear patterns. PCA changes the feature space, and its effect may differ between classifiers.

The number of principal components needed to retain 95% of variance indicates how much the input space can be compressed under this threshold. The printed scores show whether compression comes with a performance cost. Results should be reported as generated and should not be assumed to generalize to every dataset.

Limitations

- Only one public dataset and one train-test split are used.
- The dataset may not represent all patient populations or clinical settings.
- Default model configurations are used rather than exhaustive tuning.
- PCA preserves variance, not necessarily the most discriminative information.
- The results are educational and cannot diagnose an individual.
- A held-out test subset does not replace external validation.

Ethical Considerations

Healthcare classification can affect important decisions. A model may perform differently across populations or data-collection settings. Dataset origin should be documented, class-specific errors should be evaluated, and performance claims should be limited to the evidence. Qualified professionals must assess any future clinical application.

8. Conclusion and Future Scope

Conclusion

This research presents a machine learning workflow for classifying breast tumors as benign or malignant using numerical diagnostic features. The data is divided into stratified training and test sets. StandardScaler is fitted on training features, and PCA retains 95% of explained variance. Logistic Regression, KNN, and Random Forest are evaluated on original standardized features and the reduced feature representation.

Accuracy, precision, recall, F1-score, and confusion matrices are used to compare the models. The program prints measured results and the number of PCA components, allowing the effect of feature reduction to be evaluated from evidence. Whether PCA is beneficial depends on the model and the observed test results; it should not be assumed in advance.

Future Scope

1. Perform repeated stratified cross-validation for more stable estimates.
2. Tune model parameters using cross-validation without using the final test set.
3. Compare PCA thresholds such as 90%, 95%, and 99%.
4. Evaluate ROC-AUC and precision-recall curves.
5. Study model interpretability and feature importance using appropriate methods.
6. Validate on independent datasets before considering clinical use.

9. References

1. Wolberg, W. H., Street, W. N., and Mangasarian, O. L. Wisconsin Diagnostic Breast Cancer dataset, UCI Machine Learning Repository.
2. Scikit-learn Developers. Breast cancer dataset: https://scikit-learn.org/stable/modules/generated/sklearn.datasets.load_breast_cancer.html
3. Scikit-learn Developers. PCA: https://scikit-learn.org/stable/modules/generated/sklearn.decomposition.PCA.html
4. Scikit-learn Developers. StandardScaler: https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.StandardScaler.html
5. Scikit-learn Developers. LogisticRegression: https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LogisticRegression.html
6. Scikit-learn Developers. KNeighborsClassifier: https://scikit-learn.org/stable/modules/generated/sklearn.neighbors.KNeighborsClassifier.html
7. Scikit-learn Developers. RandomForestClassifier: https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html
8. Scikit-learn Developers. Classification metrics: https://scikit-learn.org/stable/modules/model_evaluation.html

Note: This paper is for academic learning and is not a validated clinical decision system.